Functions And Graphs

Content

Solution 2

1. The modulus function

Solve $|3x - 5| = 2 - \frac{1}{2}x$.

2. Mappings, domain and range

If $f(x) = x^2 - 4x + 3$, find the range of f.

3. Composite functions

For $f(x) = 2x + 1$, $g(x) = x^2$, determine $fg(x)$ and $gf(x)$.

4. Inverse functions

If $f(x) = \dfrac{x+1}{2x-1}$, determine $f^{-1}(x)$.

5. Transforming graphs

Sketch $y = |f(x)|$ and $y = f(|x|)$.

Combine transformations and transform the modulus function.

The Modulus Function

Solution 3

The modulus of a number

The modulus of a, written $|a|$, is its non-negative numerical value.

$|6| = 6, \qquad |-7.1| = 7.1.$

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Distance and magnitude

The distance between two numbers a and b is $|b - a|$, regardless of their order.

For example, the distance between 2 and 7 is

$|7 - 2| = |2 - 7| = 5.$

Modulus measures size: distances and speeds are non-negative. The same notation $|\mathbf{a}|$ gives the magnitude (length) of a vector.

Examples

Solution 4

If $f(x) = |2x - 3| + 1$, find:

(a) $f(5)$

a

$$f(5) = |10 - 3| + 1 = 7 + 1 = 8.$$

(b) $f(-2)$

b

$$f(-2) = |-4 - 3| + 1 = 7 + 1 = 8.$$

(c) $f(1)$

c

$$f(1) = |2 - 3| + 1 = 1 + 1 = 2.$$

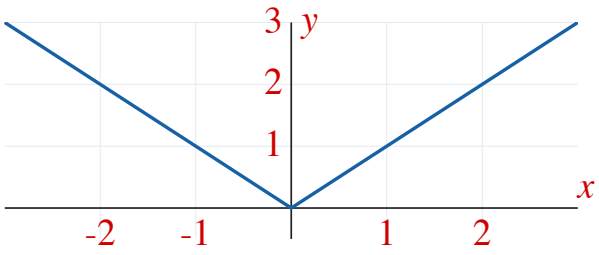
Modulus Graphs — Solution 5

Sketch $y = |x|$

Complete the values, then sketch the graph.

$x = -2$	$x = -1$	$x = 0$	$x = 1$	$x = 2$
$y = 2$	$y = 1$	$y = 0$	$y = 1$	$y = 2$

Graph



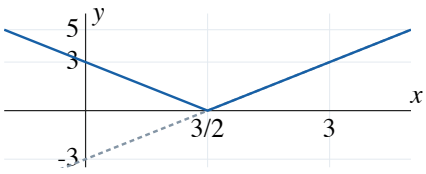
How to sketch $y = |ax + b|$

Sketch $y = ax + b$. Keep the part on or above the x-axis and reflect the part below the axis upwards.

Every output of a modulus function is non-negative.

Modulus Graphs — Solution 6

1. Sketch $y = |2x - 3|$



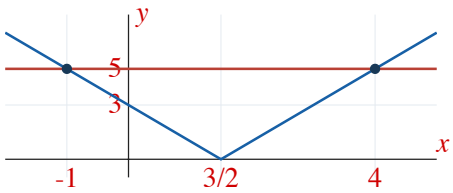
The dotted line is $y = 2x - 3$.

Reflect its negative part upwards.

The vertex is $(\frac{3}{2}, 0)$.

2. Solve $|2x - 3| = 5$

2: Graph



2: Solutions

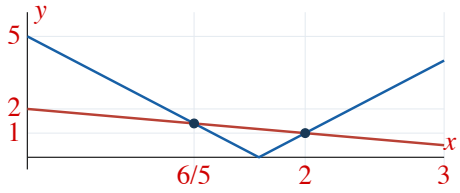
$2x - 3 = 5 \Rightarrow x = 4,$

$3 - 2x = 5 \Rightarrow x = -1.$

Both values are valid.

3. Solve $|3x - 5| = 2 - \frac{1}{2}x$

3: Graph



3: Solutions

$3x - 5 = 2 - \frac{1}{2}x \Rightarrow x = 2,$

$5 - 3x = 2 - \frac{1}{2}x \Rightarrow x = \frac{6}{5}.$

4. Solve $|3x - 5| > 2 - \frac{1}{2}x$

4

The modulus graph lies above the line outside the intersections:

$$x < \frac{6}{5} \quad \text{or} \quad x > 2.$$

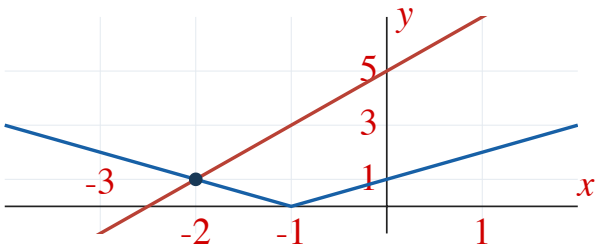
Strict inequality excludes both endpoints.

Test Your Understanding

Solution 7

1. Solve $|x+1| = 2x+5$. There is only one solution.

1: Graph



1: Working

For $x \geq -1$, $x+1 = 2x+5$ gives $x = -4$, outside this branch, so reject it.

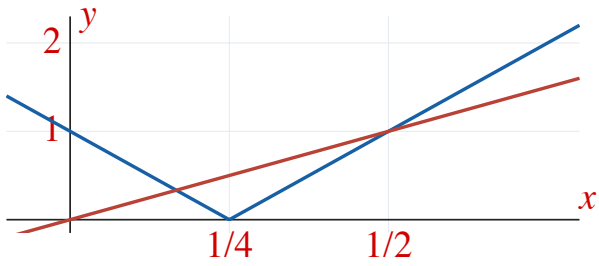
For $x < -1$:

$-x-1 = 2x+5 \Rightarrow x = -2$.

Check: $|-2+1| = 1 = 2(-2)+5$.

2. Solve $|4x-1| < 2x$.

2: Graph



2: Working

The boundary equations are

$4x-1 = 2x \Rightarrow x = \frac{1}{2}$,

$1-4x = 2x \Rightarrow x = \frac{1}{6}$.

The blue graph is below the red line between the intersections:

$\frac{1}{6} < x < \frac{1}{2}$.

Exercise 2A – Y2 Page 26

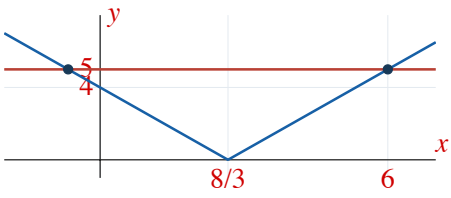
Solution 8

5. $g(x) = |4 - \frac{3}{2}x|$, $h(x) = 5$.

(a) Sketch both graphs on the same axes.

(b) Hence solve $|4 - \frac{3}{2}x| = 5$.

5a



5b

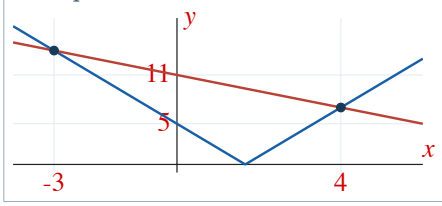
$4 - \frac{3}{2}x = 5 \Rightarrow x = -\frac{2}{3}$,

$\frac{3}{2}x - 4 = 5 \Rightarrow x = 6$.

The vertex is $(\frac{8}{3}, 0)$; the vertical intercept is 4.

8. Solve $|3x-5| = 11 - x$.

8: Graph



8

$5 - 3x = 11 - x \Rightarrow x = -3$,

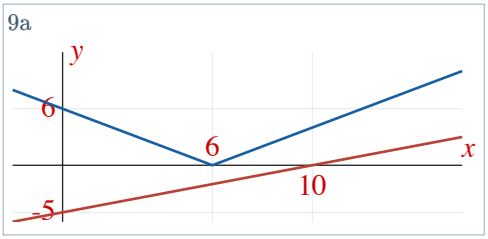
$3x - 5 = 11 - x \Rightarrow x = 4$.

Both values satisfy the original equation.

9. (a) On the same axes, sketch $y = |6-x|$ and $y = \frac{1}{2}x - 5$.

(b) State, with a reason, whether $|6-x| = \frac{1}{2}x - 5$ has any solutions.

9a



9b

There are no intersections, so no solutions.

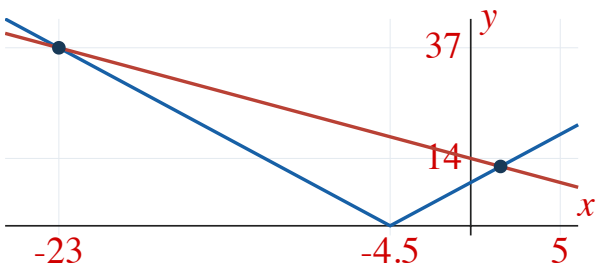
For $x \leq 6$, the equation gives $x = \frac{22}{3} > 6$, outside that branch.

For $x \geq 6$, it gives $x = 2 < 6$, again outside the branch.

Exercise 2A – Y2 Page 26 — Solution 9

12. Solve $|2x + 9| < 14 - x$.

12: Graph



12

$-2x - 9 = 14 - x \Rightarrow x = -23,$

$2x + 9 = 14 - x \Rightarrow x = \frac{5}{3}.$

The modulus graph lies below the line between the intersections:

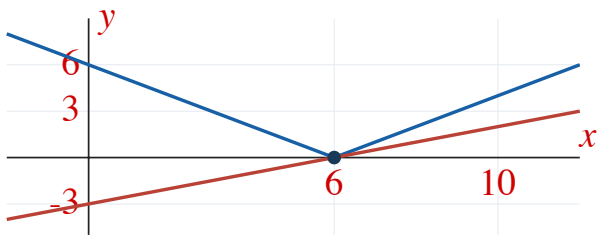
$-23 < x < \frac{5}{3}.$

13. The equation $|6 - x| = \frac{1}{2}x + k$ has exactly one solution.

(a) Find k. (2 marks)

(b) State the solution. (2 marks)

13: Graph



13a

Exactly one intersection occurs at the vertex (6, 0):

0 = ½(6) + k, so k = −3.

13b

$x = 6.$

What is a mapping? — Solution 10

What is a mapping?

A mapping relates a set of inputs to a set of outputs. It may follow a rule or be described by individual arrows.

The **domain** is the set of possible inputs. The **range** is the set of outputs that are actually reached.

An arbitrary mapping

$0 \mapsto 4.4$ and 7.2, $\quad 2 \mapsto 1$, $\quad 3.1 \mapsto 7.2$.

One input can relate to several outputs; different inputs can share an output. Some listed inputs may have no arrow.

Example: $x \mapsto 2x$

x	-1	0	1.7	2
$2x$	-2	0	3.4	4

A general relation may have more than one output for an input, or no output for some listed inputs. A function has a stricter rule, explored next.

What is a function?

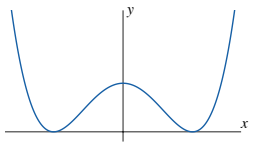
Solution 11

A function maps every element of its domain to exactly one output.

Notation: $f(x) = 2x + 1$ or $f : x \mapsto 2x + 1$. Here $f(x)$ denotes the output for input x.

A graph represents the points $(x, f(x))$. The vertical line test checks whether an input has more than one output.

A. This graph:



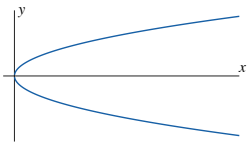
Yes

No

A: Reason

Every vertical line meets the graph at most once.

B. This graph:



Yes

No

B: Reason

A positive input gives two outputs.

C. $f(x) = 2^x$

Domain: $\mathbb{R}$.

Yes

No

C: Reason

Every real input has exactly one real output.

D. $f(x) = \sqrt{x}$

Domain: $\mathbb{R}$.

Yes

No

D: Reason

Negative inputs have no real output, so this domain is not valid.

E. $f(x) = \pm\sqrt{x}$

Domain: $x \geq 0$.

Yes

No

E: Reason

For example, input 4 gives both 2 and -2.

One-to-one vs Many-to-one

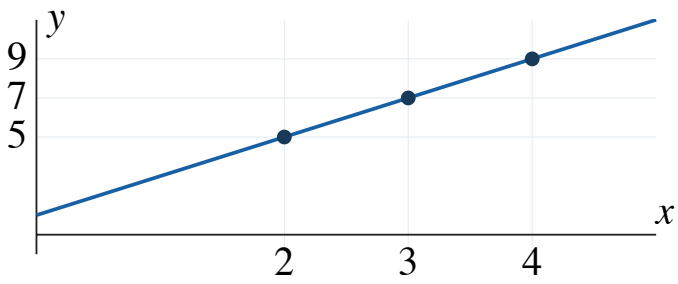
Solution 12

Each input of a function has one output. Different inputs may still share the same output. The horizontal line test distinguishes these two types.

One-to-one

Every output in the range comes from exactly one input.

Example: $f(x) = 2x + 1$.



One-to-one

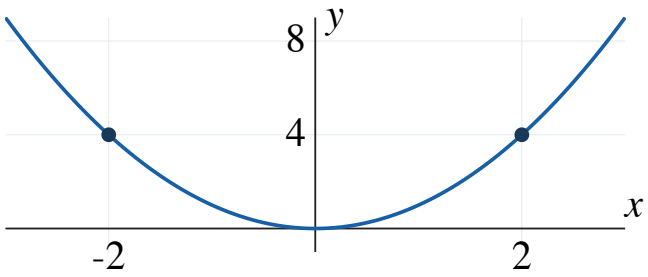
$2 \mapsto 5, \quad 3 \mapsto 7, \quad 4 \mapsto 9.$

No horizontal line intersects the graph more than once.

Many-to-one

Two or more inputs can have the same output.

Example: $f(x) = x^2$, $x \in \mathbb{R}$.



Many-to-one

$f(2) = 4$ and $f(-2) = 4$.

The horizontal line $y = 4$ intersects twice.

Further Examples

Solution 13

Find each range and state whether the function is one-to-one or many-to-one. Use a sketch where helpful. Write ranges in terms of the output, not the input.

(a) $f(x) = 3x - 2$
Domain: $\{1, 2, 3, 4\}$.

(b) $g(x) = x^2$
Domain: $-5 \le x \le 5$.

(c) $h(x) = 1/x$
Domain: $0 < x \le 3$.

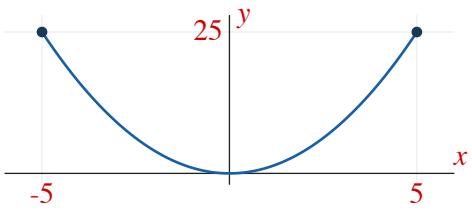
a

x	1	2	3	4
$f(x)$	1	4	7	10

Range: $\{1, 4, 7, 10\}$.

One-to-one.

b: Sketch

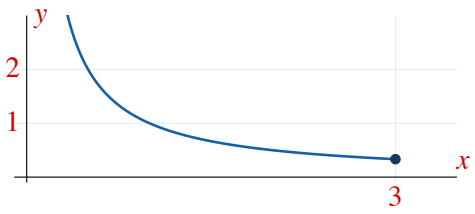


b: Range

Range: $0 \le g(x) \le 25$.

Many-to-one.

c: Sketch



c: Range

Range: $h(x) \ge \frac{1}{3}$.

One-to-one.

Piecewise Functions

Solution 14

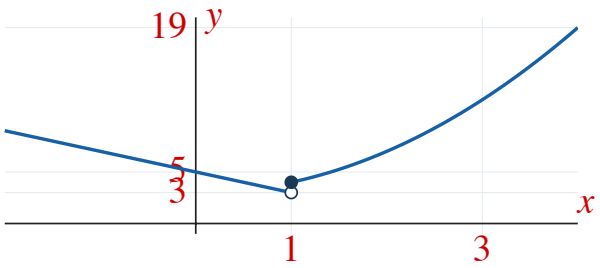
A **piecewise function** uses different rules on different parts of its domain.

$$f(x) = \begin{cases} 5 - 2x, & x < 1, \\ x^2 + 3, & x \ge 1. \end{cases}$$

(a) Sketch $y = f(x)$ and state the range.

(b) Solve $f(x) = 19$.

a: Sketch



a: Range

The left branch approaches 3 but does not reach it. The right branch starts at 4. Together they give $f(x) > 3$.

An open circle excludes an endpoint; a filled circle includes it.

b

For $x < 1$: $5 - 2x = 19 \Rightarrow x = -7$.

For $x \ge 1$: $x^2 + 3 = 19 \Rightarrow x = \pm 4$. Only $x = 4$ is in this branch.

Therefore $x = -7$ or $x = 4$.

Reject $x = -4$: it belongs to the unused part of the quadratic.

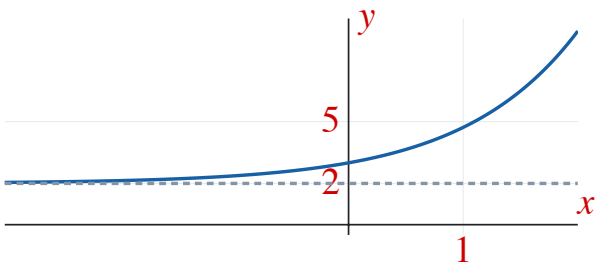
Test Your Understanding

Solution 15

Edexcel C4 June 2012, Q6(a)

$f(x) = e^x + 2$, $x \in \mathbb{R}$. State the range of f.

1: Sketch



1: Range

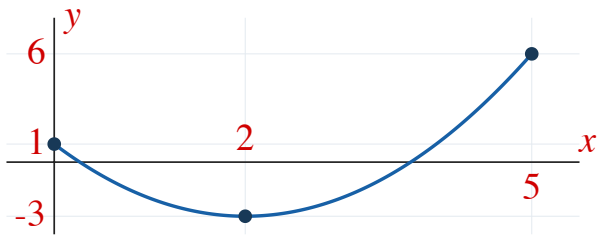
Since $e^x > 0$, the range is $f(x) > 2$.

The asymptote $y = 2$ is never reached.

Edexcel C4 June 2010, Q4(d)

$g(x) = x^2 - 4x + 1$, $0 \le x \le 5$. Find the range.

2: Sketch



2: Range

$g(x) = (x-2)^2 - 3$, so the minimum is -3 at $x = 2$.

At the domain endpoints, $g(0) = 1$ and $g(5) = 6$.

Hence $-3 \le g(x) \le 6$.

Exercise 2B – Y2 Pages 30-32

Solution 16

7. The function s is defined by

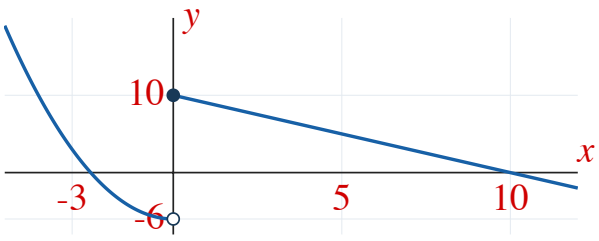
$$s(x) = \begin{cases} x^2 - 6, & x < 0, \\ 10 - x, & x \ge 0. \end{cases}$$

(a) Sketch $y = s(x)$.

(b) Find all a such that $s(a) = 43$.

(c) Solve $s(x) = x$.

7a



7b

For $a < 0$: $a^2 - 6 = 43 \Rightarrow a = \pm 7$; retain $a = -7$.

For $a \ge 0$: $10 - a = 43 \Rightarrow a = -33$, outside this branch.

The only solution is $a = -7$.

7c

For $x < 0$:

$x^2 - 6 = x \Rightarrow (x-3)(x+2) = 0$.

Retain $x = -2$.

For $x \ge 0$: $10 - x = x \Rightarrow x = 5$.

Therefore $x = -2$ or $x = 5$.

Exercise 2B – Y2 Pages 30-32 — Solution 17

8. The function p is defined by

$$p(x) = \begin{cases} e^{-x}, & -5 \le x < 0, \\ x^3 + 4, & 0 \le x \le 4. \end{cases}$$

(a) Sketch $y = p(x)$.

(b) Find all a, to 2 decimal places, such that $p(a) = 50$.

8a



The left branch ends at the open point $(0, 1)$; the right branch starts at the filled point $(0, 4)$.

8b

On the negative branch:

$e^{-a} = 50 \Rightarrow -a = \ln 50 \Rightarrow a = -3.91.$

On the non-negative branch:

$a^3 + 4 = 50 \Rightarrow a = \sqrt[3]{46} = 3.58.$

Both values lie within their branch domains.

The solutions are $a = -3.91$ and $a = 3.58$, to 2 decimal places.

Exercise 2B – Y2 Pages 30-32 — Solution 18

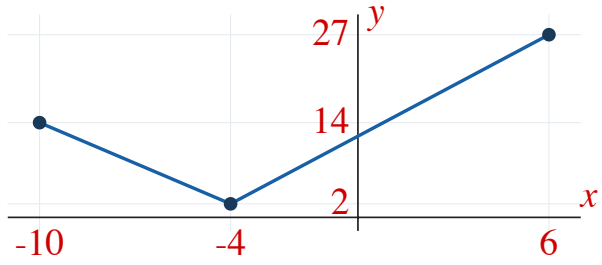
9. The function h has domain $-10 \le x \le 6$. It is linear from $(-10, 14)$ to $(-4, 2)$, and from $(-4, 2)$ to $(6, 27)$.

(a) Sketch $y = h(x)$.

(b) State the range.

(c) Find all a such that $h(a) = 12$.

9a



9b

The lowest output is 2 and the greatest is 27.

$2 \le h(x) \le 27.$

9c

The first segment has slope -2, so $h(x) = -2x - 6$.

$-2a - 6 = 12 \Rightarrow a = -9.$

The second has slope $\frac{5}{2}$, so $h(x) = \frac{5}{2}x + 12$.

$\frac{5}{2}a + 12 = 12 \Rightarrow a = 0.$

Both lie in the corresponding intervals.

Exercise 2B – Y2 Pages 30-32 Solution 19

11. $f(x) = ax^3 + bx - 5$, where a, b are constants. Given $f(1) = -4$ and $f(2) = 9$, find a, b.

11

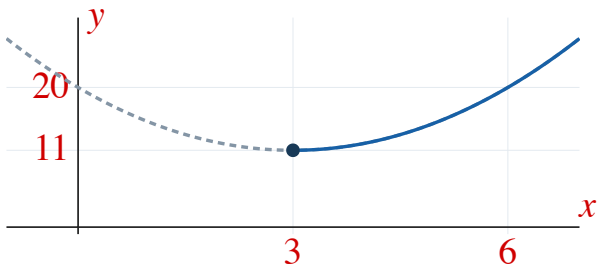
$a + b - 5 = -4 \Rightarrow a + b = 1,$

$8a + 2b - 5 = 9 \Rightarrow 4a + b = 7.$

Subtracting gives $3a = 6$, so $a = 2$. Then $2 + b = 1$, so $b = -1$.

12. $h(x) = x^2 - 6x + 20$, with domain $x \geq a$. Given that h is one-to-one, find the smallest possible a. (6 marks)

12: Sketch



12

$h(x) = (x - 3)^2 + 11.$

The vertex is $(3, 11)$. The function is one-to-one on the increasing branch $x \geq 3$. Any smaller lower bound includes two inputs with the same output.

Therefore the smallest possible value is $a = 3$.

Summary of Domain/Range Solution 20

Identify the range of each function. Sketch the graph and check whether the domain is restricted.

1. $f(x) = x^2, x \in \mathbb{R}$.

1

$f(x) \geq 0.$

3. $f(x) = \ln x, x > 0$.

3

$f(x) \in \mathbb{R}.$

5. $f(x) = x^2 + 2x + 9, x \in \mathbb{R}$.

5

$f(x) = (x + 1)^2 + 8$, so $f(x) \geq 8.$

2. $f(x) = 1/x, x \neq 0$.

2

$f(x) \neq 0.$

4. $f(x) = e^x, x \in \mathbb{R}$.

4

$f(x) > 0.$

6. $f(x) = x^2, -1 \leq x \leq 4$.

6

$0 \leq f(x) \leq 16.$

Just for your interest... Solution 21

Function or output?

Compare $f : x \mapsto 2x + 1$ and $f(x) = 2x + 1$.

The first describes the mapping f; the second gives its output for input x. A variable can represent a function, just as it can represent a number, vector or set.

Functional equations

Solving $2x + 1 = 5$ finds a number, $x = 2$. Solving

$f(x + y) = f(x)f(y)$

means finding functions that satisfy it.

Example

One example is $f(x) = 2^x$, because

$2^{x+y} = 2^x 2^y$.

This verifies one solution; it does not find every possible function.

Functions as inputs

A computer program can accept a function as an input. For example, applying $x \mapsto x + 1$ to every entry of $[1, 2, 3]$ gives $[2, 3, 4]$.

A map operation builds a new list by applying f to each element of the original list.

Composite Functions Solution 22

Composite functions

Applying functions in succession creates a composite function.

$x \xrightarrow{f} f(x) \xrightarrow{g} g(f(x))$

$gf(x) = g(f(x))$: apply f first, then g.

Read a composition from right to left. The order generally matters.

Example

If $f(x) = x + 1$ and $g(x) = 2x$, compare $gf(x)$ and $fg(x)$.

Compare

$gf(x) = 2(x + 1) = 2x + 2.$

$fg(x) = 2x + 1.$

They are different functions.

Examples

Solution 23

Let $f(x) = x^2 + 1$ and $g(x) = 4x - 2$. Replace each input of the outer function with the whole inner expression.

(a) Find $fg(2)$.

a $fg(2) = f(6) = 6^2 + 1 = 37.$

(b) Find $fg(x)$.

b
$$\begin{aligned} fg(x) &= f(4x-2) \\ &= (4x-2)^2 + 1 \\ &= 16x^2 - 16x + 5. \end{aligned}$$

(c) Find $gf(x)$.

c
$$\begin{aligned} gf(x) &= g(x^2+1) \\ &= 4(x^2+1) - 2 \\ &= 4x^2 + 2. \end{aligned}$$

(d) Find $f^2(x)$.

d
$f^2(x) = f(f(x)) = (x^2+1)^2 + 1.$
Here f^2 means apply f twice, not square its output.

(e) Solve $gf(x) = 38$.

e
$$\begin{aligned} 4x^2 + 2 &= 38, \\ x^2 &= 9, \\ x &= \pm 3. \end{aligned}$$

Further Examples

Solution 24

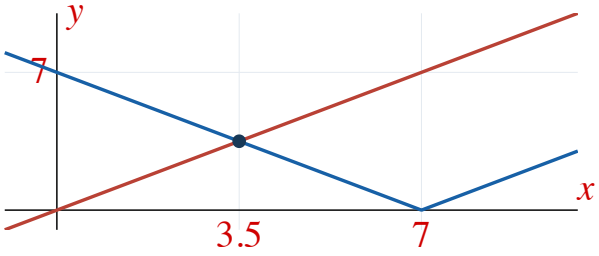
$f(x) = |2x - 8|$, $g(x) = \dfrac{x+1}{2}$.

(a) Find $fg(3)$.

(b) Solve $fg(x) = x$.

a $fg(3) = f(2) = |4 - 8| = 4.$

b: Sketch



b: Working

First form the composite:

$$fg(x) = \left|2\left(\frac{x+1}{2}\right) - 8\right| = |x - 7|.$$

For $x < 7$: $7 - x = x \Rightarrow x = \frac{7}{2}$.

For $x \geq 7$: $x - 7 = x$ is impossible.

The only solution is $x = 3.5$.

Test Your Understanding — Solution 25

Edexcel C4 June 2013(R), Q4

$f(x) = 2|x| + 3$, $g(x) = 3 - 4x$, both on $\mathbb{R}$.

(b) Find $fg(1)$.

(d) Solve $gg(x) + [g(x)]^2 = 0$.

b $fg(1) = f(-1) = 2|-1| + 3 = 5.$

d

$$3 - 4(3 - 4x) + (3 - 4x)^2 = 0,$$
$$16x^2 - 8x = 0,$$
$$8x(2x - 1) = 0.$$

Hence $x = 0$ or $x = \frac{1}{2}$.

Edexcel C4 June 2012, Q6(b)

$f(x) = e^x + 2$, $x \in \mathbb{R}$, and $g(x) = \ln x$, $x > 0$.

Find $fg(x)$ in its simplest form.

Answer

$fg(x) = e^{\ln x} + 2 = x + 2, \qquad x > 0.$

Exponential and natural logarithm are inverse operations. The composite retains the restriction $x > 0$.

Exercise 2C – Y2 Pages 34-35 — Solution 26

3. $f(x) = 3x - 2$, $g(x) = x^2$, $x \in \mathbb{R}$.

(a) Find $fg(x)$.

(b) Solve $fg(x) = gf(x)$.

3a $fg(x) = 3x^2 - 2.$

3b

$$3x^2 - 2 = (3x - 2)^2,$$
$$6x^2 - 12x + 6 = 0,$$
$$(x - 1)^2 = 0.$$

Therefore $x = 1$.

4. $p(x) = \dfrac{1}{x - 2}$, $x \neq 2$, and $q(x) = 3x + 4$, $x \in \mathbb{R}$.

(a) Express $qp(x)$ as $\dfrac{ax + b}{cx + d}$.

(b) Solve $qp(x) = 16$.

4a $qp(x) = \dfrac{3}{x - 2} + 4 = \dfrac{4x - 5}{x - 2}, \qquad x \neq 2.$

4b

$$\frac{3}{x - 2} + 4 = 16 \Rightarrow x - 2 = \frac{1}{4}.$$

Hence $x = \frac{9}{4}$.

Exercise 2C – Y2 Pages 34-35

Solution 27

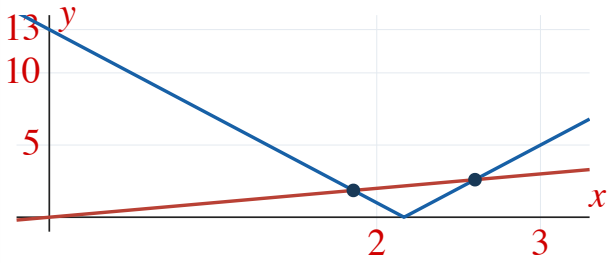
5. $f(x) = |9 - 4x|$, $g(x) = \dfrac{3x-2}{2}$.

(a) Find $fg(6)$.

(b) Solve $fg(x) = x$.

5a $g(6) = 8, \qquad fg(6) = |9 - 32| = 23.$

5b: Sketch



5b: Working

$$fg(x) = \left|9 - 4\left(\frac{3x-2}{2}\right)\right| = |13 - 6x|.$$

Solve each branch:

$$13 - 6x = x \Rightarrow x = \frac{13}{7},$$

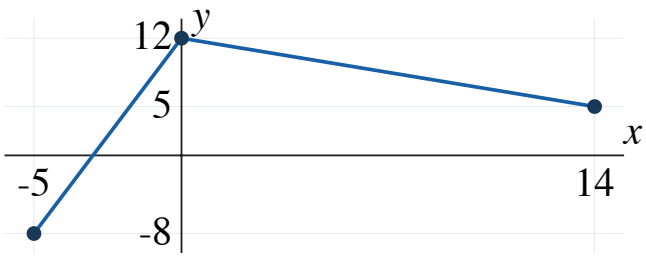
$$6x - 13 = x \Rightarrow x = \frac{13}{5}.$$

Both values satisfy $fg(x) = x$.

Exercise 2C – Y2 Pages 34-35

Solution 28

11. The domain of g is $-5 \le x \le 14$. It is linear from $(-5, -8)$ to $(0, 12)$, then to $(14, 5)$.



(a) State the range of g.

(b) Find $gg(0)$.

(c) Given $h(x) = \dfrac{2x-5}{10-x}$, find $gh(7)$.

11a

$-8 \le g(x) \le 12.$

11b

For $0 \le x \le 14$, $g(x) = 12 - \frac{1}{2}x$.

$g(0) = 12, \qquad gg(0) = g(12) = 6.$

11c

$$h(7) = \frac{14-5}{10-7} = 3.$$

$$gh(7) = g(3) = 12 - \frac{3}{2} = \frac{21}{2}.$$

EXTENSION

Solution 29

MAT 2014, 1F

$S(x) = x + 1$, $T(x) = -x$. Apply S a total of s times and T a total of t times, in some order, to obtain $8 - x$. What follows?

(i) $s = 8, t = 1$; (ii) s odd, t even; (iii) s even, t odd; (iv) both powers of 2; (v) none.

Each T reverses the coefficient of x, so t is odd.
Each S changes the constant's parity; T preserves it. The final constant is even, so s is even.

The counts are not unique. Therefore **(iii)**.

MAT 2012, Q2

$f(x) = x + 1$, $g(x) = 2x$.

(i) Show $f^2g(x) = gf(x)$.

(ii) Find all compositions giving $4x + 4$.

(iii) Find $f^i g f^j g f^k(x)$, for integers $i, j, k \geq 0$.

(iv) How many compositions give $4x + 4m$, where $m \geq 0$ is an integer?

i

(i) Both equal $2x + 2$.

(ii) $f^4g^2,\ f^2gfg,\ gf^2g,\ g^2f$.

(iii) $4x + 4k + 2j + i$.

(iv) Require $4k + 2j + i = 4m$. For each $k = 0, \ldots, m$, there are $2(m - k) + 1$ choices of j, with i then fixed.

$1 + 3 + \cdots + (2m + 1) = (m + 1)^2$.

Inverse Functions

Solution 30

Undoing a function

An inverse function reverses the original mapping. If $f(4) = 2$, then $f^{-1}(2) = 4$.

For $f(x) = 2x + 1$:

$x \xrightarrow{\times 2} 2x \xrightarrow{+1} 2x + 1$.

Undo the operations in reverse order: subtract 1, then divide by 2.

$$f^{-1}(x) = \frac{x - 1}{2}.$$

Why must f be one-to-one?

Reason

If two different inputs had the same output, reversing the arrows would give two outputs for one input. That would not be a function.

Notation

f^{-1} means the inverse function; it does not mean $1/f$.

Similarly, $\sin^{-1}$ denotes inverse sine.

Applying f, then f^{-1}, returns the original input.

More on Inverse Functions

Solution 31

Method

Write $y = f(x)$, make x the subject, then exchange the variable names.

Example: $y = 2x + 1$ gives $x = \dfrac{y-1}{2}$, hence $f^{-1}(x) = \dfrac{x-1}{2}$.

1. Find the inverse of $f(x) = 3 - 4x$.

1

$y = 3 - 4x,$

$4x = 3 - y,$

$x = \dfrac{3-y}{4}.$

Thus $f^{-1}(x) = \dfrac{3-x}{4}.$

2. Find the inverse of $f(x) = \dfrac{x+2}{2x-1}$, $x \neq \frac{1}{2}$.

2

$y(2x-1) = x + 2,$

$x(2y-1) = y + 2,$

$x = \dfrac{y+2}{2y-1}.$

Therefore $f^{-1}(x) = \dfrac{x+2}{2x-1}$, $x \neq \frac{1}{2}$. This function is its own inverse.

Graphing an Inverse Function

Solution 32

Reflect in $y = x$

Inverting swaps input and output: (a, b) on $y = f(x)$ becomes (b, a) on $y = f^{-1}(x)$.

The graphs are reflections in $y = x$. Vertical intercepts become horizontal intercepts; vertical asymptotes become horizontal ones.

The domain of f is the range of f^{-1}, and vice versa.

Domain → range

If the domain of f is $x > 1$, the range of f^{-1} is $f^{-1}(x) > 1$.

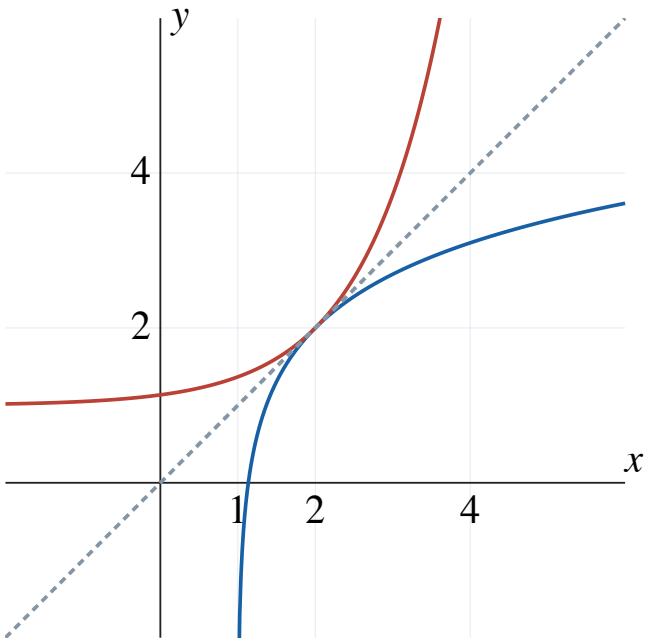


Illustration: blue $f(x) = \ln(x-1) + 2$; red $f^{-1}(x) = e^{x-2} + 1$. The dashed line is $y = x$.

Example

Solution 33

$g(x) = \sqrt{x-2}$, with domain $x \geq 2$.

(a) State the range.

(b) Find $g^{-1}(x)$.

(c) Sketch both functions.

(d) State the domain and range of g^{-1}.

a

$g(x) \geq 0$.

b

$y = \sqrt{x-2} \Rightarrow x = y^2 + 2$.

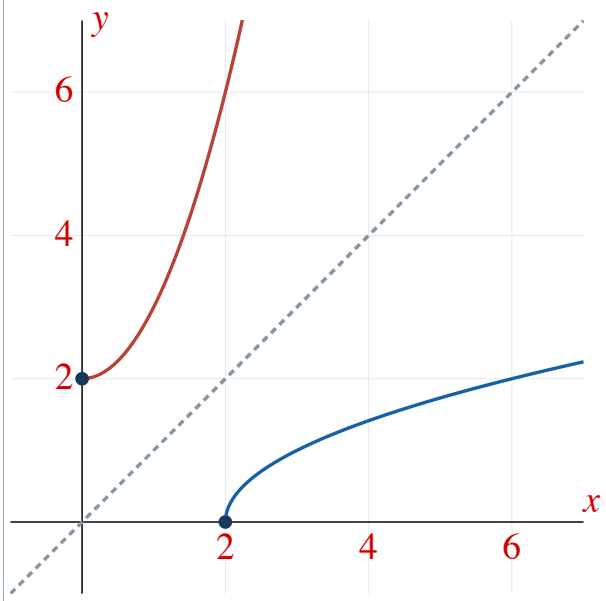
Hence $g^{-1}(x) = x^2 + 2$.

d

Domain: $x \geq 0$.

Range: $g^{-1}(x) \geq 2$.

c: Sketch



Blue: g. Red: g^{-1}. Dashed: $y = x$.

Further Example

Solution 34

$f(x) = x^2 - 3$, with domain $x \geq 0$.

(a) Find f^{-1}.

(b) Sketch both functions and state the domain of f^{-1}.

(c) Solve $f(x) = f^{-1}(x)$.

a / b: Domain

$y = x^2 - 3 \Rightarrow x = \sqrt{y+3}$.

Thus $f^{-1}(x) = \sqrt{x+3}$, with domain $x \geq -3$.

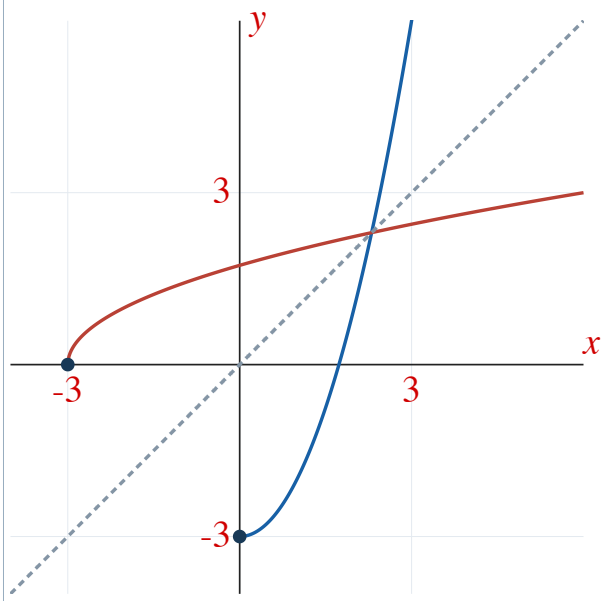
c

For this increasing function, intersections with its inverse lie on $y = x$:

$$x^2 - 3 = x \Rightarrow x = \frac{1 \pm \sqrt{13}}{2}.$$

Since $x \geq 0$, retain $x = \dfrac{1 + \sqrt{13}}{2}$.

b: Sketch



Blue: f. Red: f^{-1}. Dashed: $y = x$.

Test Your Understanding

Solution 35

Edexcel C4 June 2012, Q6

$f(x) = e^x + 2, x \in \mathbb{R}$.

(d) Find f^{-1}, stating its domain.

(e) Sketch both curves on the same axes, giving all axis intercepts.

d

$y = e^x + 2 \Rightarrow x = \ln(y - 2)$.

Thus $f^{-1}(x) = \ln(x - 2)$, with domain $x > 2$.

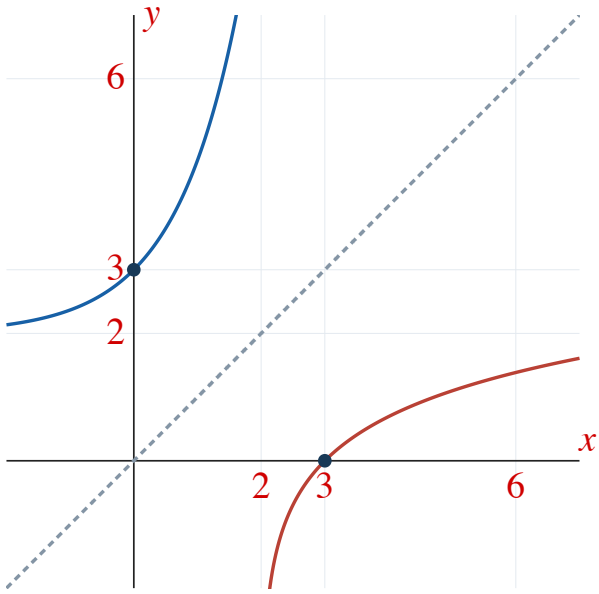
e: Features

f crosses the vertical axis at $(0, 3)$; it has no horizontal intercept.

f^{-1} crosses the horizontal axis at $(3, 0)$; it has no vertical intercept.

The asymptotes are $y = 2$ and $x = 2$, respectively.

e: Sketch



Blue: f. Red: f^{-1}. Dashed: $y = x$.

Exercise 2D – Y2 page 38-39

Solution 36

3. Explain why $g : x \mapsto 4 - x$, with domain $x > 0$, is not identical to its inverse.

Formula

$y = 4 - x \Rightarrow x = 4 - y$.

So the inverse has the same formula, $g^{-1}(x) = 4 - x$. But a function includes its domain, not just its formula.

Domains

For g: domain $x > 0$, range $g(x) < 4$.

For g^{-1}: domain $x < 4$, range $g^{-1}(x) > 0$.

The domains differ, so the functions are not identical.

For example, -1 is a valid input for g^{-1}, but not for g.

Exercise 2D – Y2 page 38-39 Solution 37

5. $t(x) = x^2 - 6x + 5$, with domain $x \geq 5$. Find $t^{-1}(x)$.

Answer

Complete the square:

$y = (x-3)^2 - 4.$

As $x \geq 5$, take the positive square root:

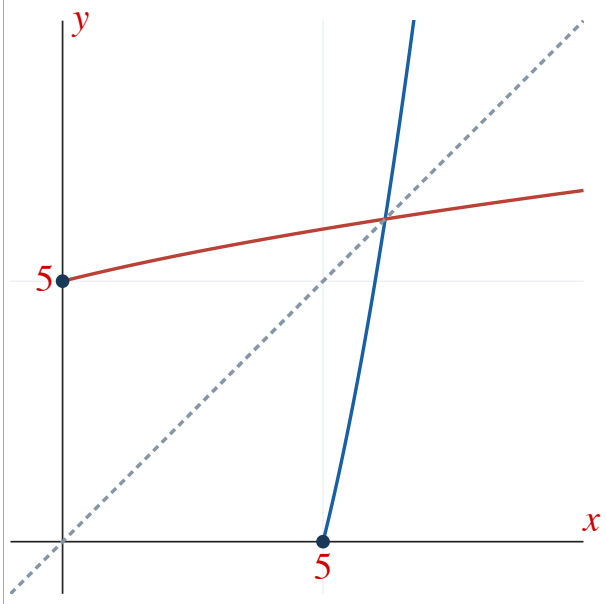
$\sqrt{y+4} = x - 3.$

Therefore

$t^{-1}(x) = 3 + \sqrt{x+4}.$

The range of t is $[0, \infty)$, so the inverse has domain $x \geq 0$.

Sketch



Blue: t. Red: t^{-1}. The endpoints $(5, 0)$ and $(0, 5)$ are included.

Exercise 2D – Y2 page 38-39 Solution 38

6. $m(x) = x^2 + 4x + 9$, with domain $x > a$, for a constant a.

(a) State the least value of a for which m^{-1} exists.

(b) Find $m^{-1}(x)$.

(c) State its domain.

6a

$m(x) = (x+2)^2 + 5.$

The vertex is $(-2, 5)$. Restricting to the right of the vertex makes the function one-to-one. The least value is $a = -2$.

6b

$$y = (x+2)^2 + 5,$$
$$\sqrt{y-5} = x + 2,$$
$$x = \sqrt{y-5} - 2.$$

Thus $m^{-1}(x) = \sqrt{x-5} - 2.$

6c

With $a = -2$, the original domain is $x > -2$, so the range is $m(x) > 5$.

The inverse therefore has domain $x > 5$. The endpoint 5 is excluded.

Exercise 2D – Y2 page 38-39 — Solution 39

10. $f(x) = 2x^2 - 3$, with domain $x < 0$.

(a) Find $f^{-1}(x)$, stating its domain.

(b) Find the values of a for which $f(a) = f^{-1}(a)$.

10a

$$y = 2x^2 - 3 \Rightarrow x = -\sqrt{\frac{y+3}{2}}.$$

Take the negative root because $x < 0$. Thus

$$f^{-1}(x) = -\sqrt{\frac{x+3}{2}}, \quad x > -3.$$

10b

Solve the original equality:

$$2a^2 - 3 = -\sqrt{\frac{a+3}{2}}.$$

It requires $-3 < a < 0$ and $2a^2 - 3 < 0$. Squaring gives

$$8a^4 - 24a^2 - a + 15 = 0,$$

$$(a+1)(2a-3)(4a^2+2a-5) = 0.$$

Candidates: -1, $\frac{3}{2}$, $\frac{-1\pm\sqrt{21}}{4}$. Only $a = -1$ satisfies the original equation: both sides equal -1.

Always check roots after squaring. For a decreasing function, intersections with its inverse need not all lie on $y = x$.

Exercise 2D – Y2 page 38-39 — Solution 40

12.

$$f(x) = \frac{3(x+2)}{x^2+x-20} - \frac{2}{x-4}, \quad x > 4.$$

(a) Show that $f(x) = \frac{1}{x+5}$.

(b) Find the range.

(c) Find $f^{-1}(x)$ and its domain.

12a

$$f(x) = \frac{3(x+2) - 2(x+5)}{(x+5)(x-4)}$$

$$= \frac{x-4}{(x+5)(x-4)}$$

$$= \frac{1}{x+5}, \quad x > 4.$$

12b

For $x > 4$, the denominator satisfies $x + 5 > 9$, so

$$0 < f(x) < \frac{1}{9}.$$

Both endpoints are excluded.

12c

$$y = \frac{1}{x+5} \Rightarrow x = \frac{1}{y} - 5.$$

Therefore

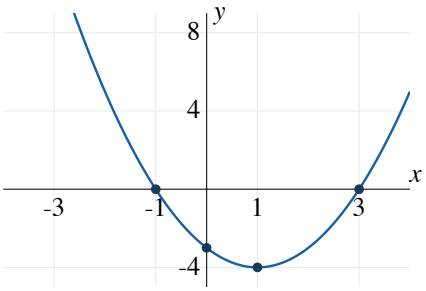
$$f^{-1}(x) = \frac{1}{x} - 5, \qquad 0 < x < \frac{1}{9}.$$

Sketching modulus transformations

Solution 41

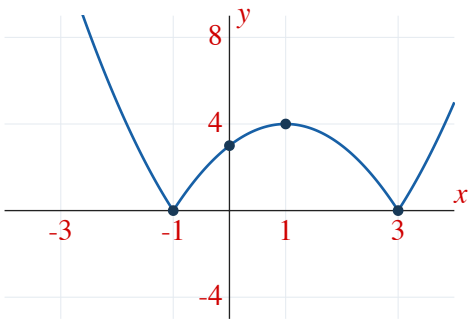
$f(x) = (x-3)(x+1)$. Sketch $y = |f(x)|$ and $y = f(|x|)$. Show the intercepts and turning points.

Given graph



(a) $y = |f(x)|$

a: Graph



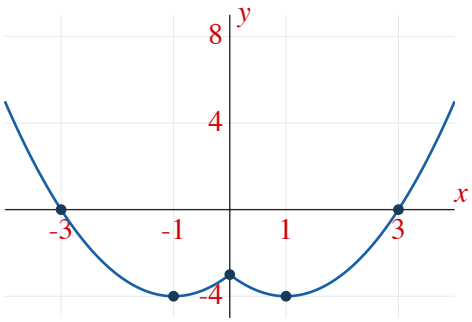
a: Details

Reflect only the parts below the horizontal axis.

Roots: $-1, 3$. Vertical intercept: 3. The turning point $(1, -4)$ becomes $(1, 4)$.

(b) $y = f(|x|)$

b: Graph



b: Details

Keep the right half and reflect it in the vertical axis.

Roots: $-3, 3$. Vertical intercept: -3. Minima: $(-1, -4)$, $(1, -4)$.

Test Your Understanding

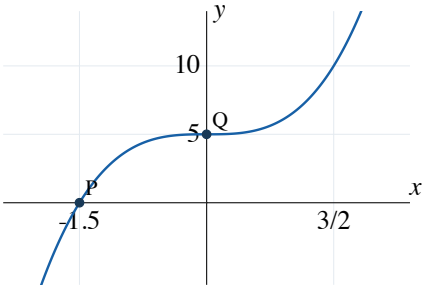
Solution 42

Edexcel C4 June 2012, Q4

The given curve passes through $P(-\frac{3}{2}, 0)$ and has a stationary point of inflection at $Q(0, 5)$. Sketch (a) $y = |f(x)|$ and (b) $y = f(|x|)$. Label all axis intercepts. (2 marks each)

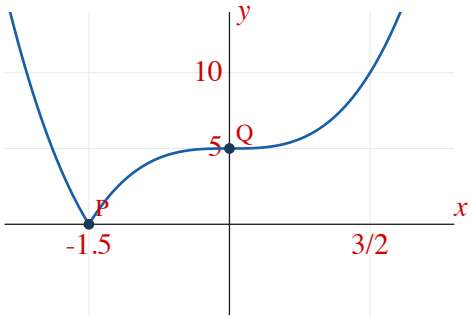
The curves below are schematic.

Given graph



(a) $y = |f(x)|$

a: Graph



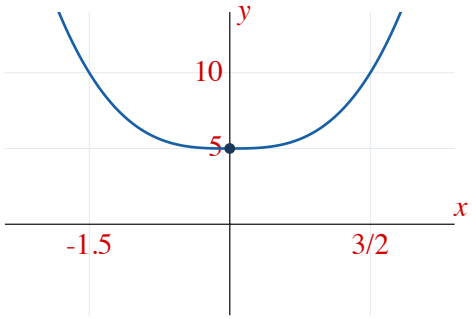
a: Details

The negative branch is reflected upwards.

Intercepts: $(-\frac{3}{2}, 0)$ and $(0, 5)$. There is a sharp point at P.

(b) $y = f(|x|)$

b: Graph



b: Details

The right-hand branch is reflected in the vertical axis.

Vertical intercept: $(0, 5)$. There are no horizontal intercepts.

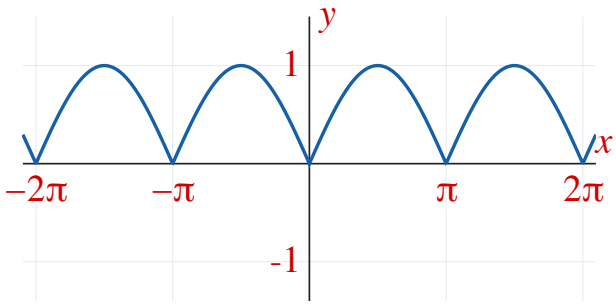
Further Test Your Understanding

Solution 43

Sketch each graph for $-2\pi \leq x \leq 2\pi$. All angles are in radians.

(a) $y = |\sin x|$

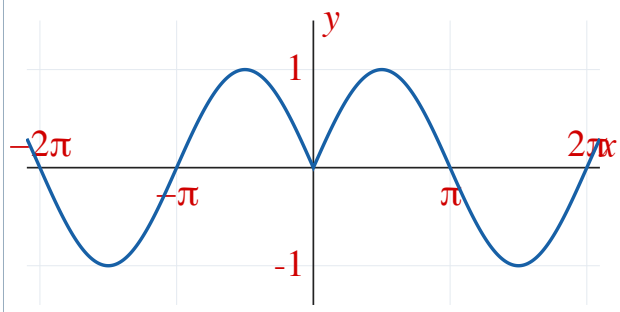
a



Reflect every negative part of $y = \sin x$ upwards. All values lie between 0 and 1.

(b) $y = \sin |x|$

b



Keep $y = \sin x$ for $x \geq 0$, then reflect this half in the vertical axis. Values are negative when $\pi < |x| < 2\pi$.

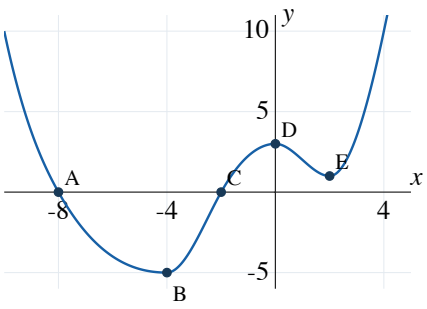
Exercise 2E – Y2 Pages 42-44

Solution 44

Exercise 2E, Q5

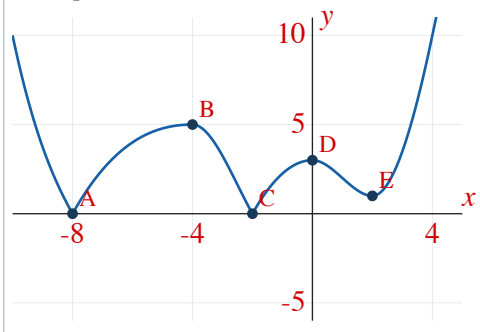
The schematic graph of f has labelled points $A(-8, 0)$, $B(-4, -5)$, $C(-2, 0)$, $D(0, 3)$, $E(2, 1)$. Sketch the two transformations, labelling corresponding points and axis intercepts. (3 marks each)

Given graph



(a) $y = |f(x)|$

a: Graph



a: Details

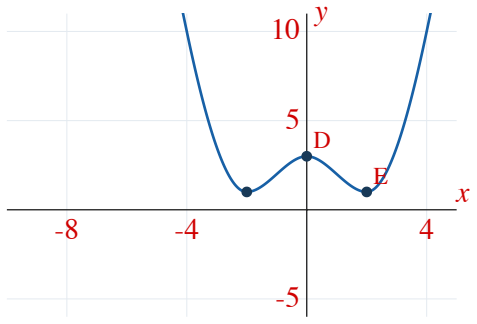
B moves to $(-4, 5)$; A, C, D, E stay in place.

Horizontal intercepts: $-8, -2$.

Vertical intercept: 3.

(b) $y = f(|x|)$

b: Graph



b: Details

Discard the original left half.

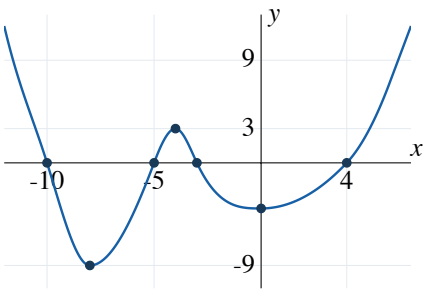
Mirror E to $(-2, 1)$; $D(0, 3)$ stays in place.

There are no horizontal intercepts.

Exercise 2E, Q6

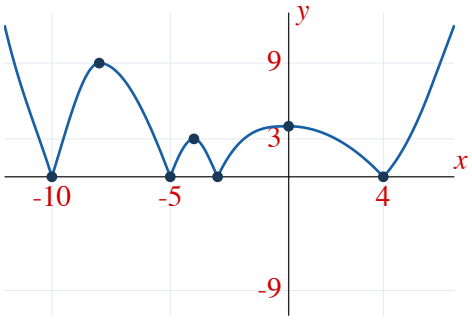
The schematic graph of f passes through $A(-10,0)$, $B(-8,-9)$, $C(-5,0)$, $D(-4,3)$, $E(-3,0)$, $F(0,-4)$, $G(4,0)$. Sketch each transformation, labelling the corresponding points and intercepts.

Given graph



(a) $y = |f(x)|$

a: Graph



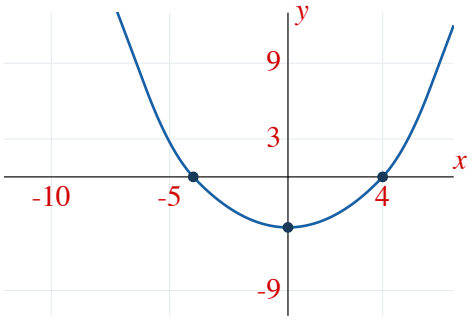
a: Details

$B \to (-8,9)$, $F \to (0,4)$. Other labelled points stay in place.

Roots: $-10, -5, -3, 4$. (4 marks)

(b) $y = f(|x|)$

b: Graph



b: Details

Keep the right half and mirror it. The minimum is $(0,-4)$; the roots are $-4, 4$. (3 marks)

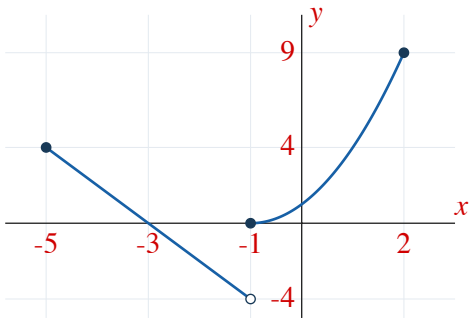
Exercise 2E, Q10

$$f(x) = \begin{cases} -2x - 6, & -5 \le x < -1, \\ (x+1)^2, & -1 \le x \le 2. \end{cases}$$

(a) Sketch f and state its range. (b) Sketch $|f(x)|$. (c) Sketch $f(|x|)$.

(a) Original function

a: Graph



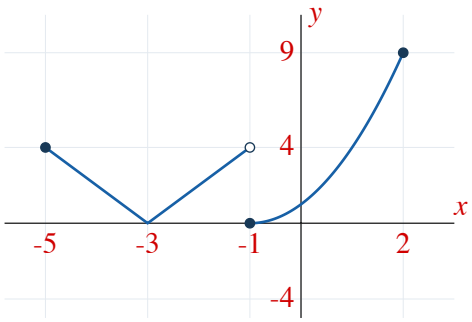
a: Range

Range: $-4 < y \le 9$.

An open circle excludes $(-1,-4)$. The point $(-1,0)$ is included.

(b) $y = |f(x)|$

b: Graph

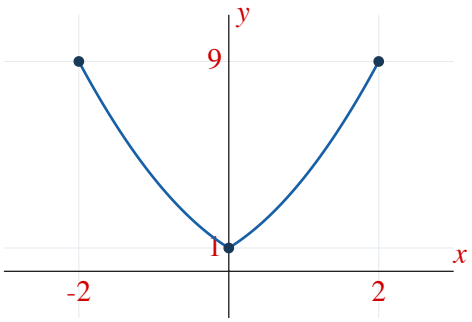


b: Details

Reflect only negative outputs. Domain: $[-5,2]$. Range: $[0,9]$.

(c) $y = f(|x|)$

c: Graph



c: Details

$y = (|x|+1)^2$, $-2 \le x \le 2$.

The minimum is $(0,1)$. Range: $[1,9]$.

EXTENSION

Solution 47

SMC 2008, Q25

Find the area of the region satisfying

$\big||x|-2\big|+\big||y|-2\big|\le 4.$

$A:24 \quad B:32 \quad C:64$

$D:96 \quad E:112$

Working

In the first quadrant, the inequality is

$|x-2|+|y-2|\le 4.$

The full diamond has diagonals 8 and 8, so area 32.

The two triangles outside the quadrant each have base 4 and height 2, so area 4.

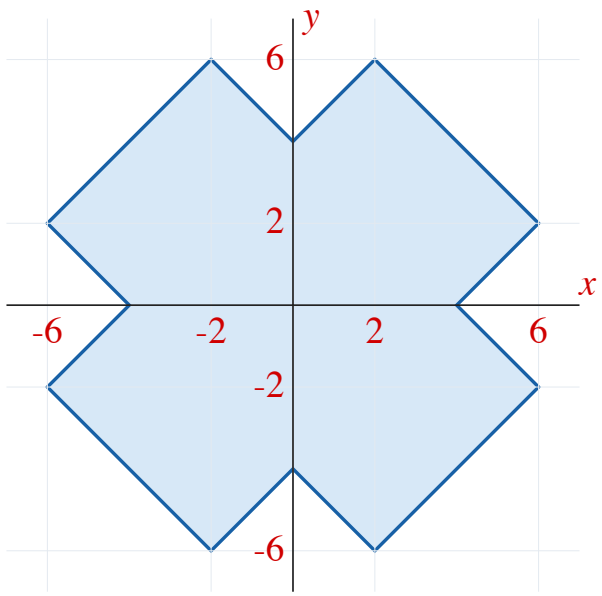
First-quadrant area: $32-4-4=24$.

By symmetry, the total area is

$4\times 24=96.$

Answer D.

Region



The boundary is included. The same area occurs in all four quadrants.

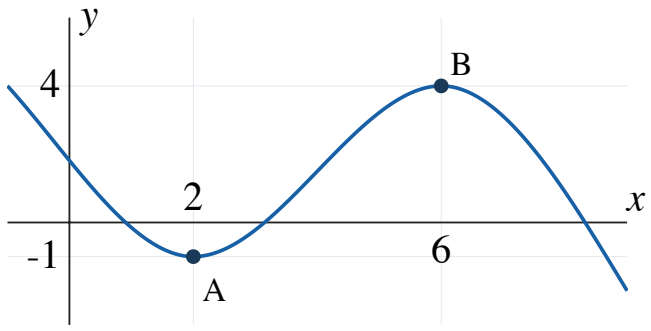
Combining Transformations

Solution 48

Combining transformations

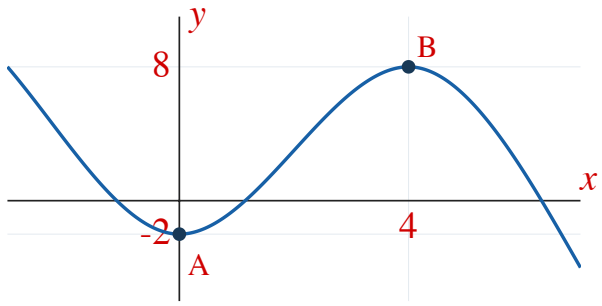
A change inside $f(\cdot)$ affects horizontal coordinates; a change outside affects vertical coordinates.

The schematic graph of $y=f(x)$ contains $A(2,-1)$ and $B(6,4)$.



(a) Sketch $y=2f(x+2)$

a: Graph



a: Coordinates

Every point maps as

$(x,y)\longmapsto(x-2,2y).$

Translate 2 left and stretch vertically by factor 2.

$A\to(0,-2), \qquad B\to(4,8).$

Start with the known points, then retain the shape between them.

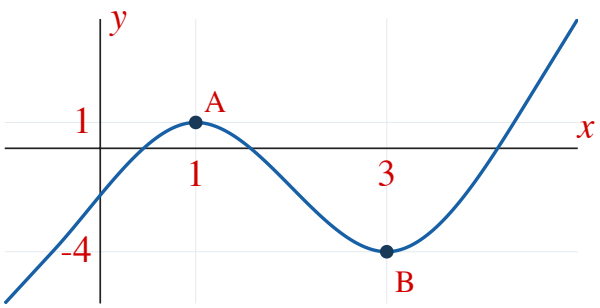
Combining Transformations

Solution 49

Continue with the graph containing $A(2,-1)$ and $B(6,4)$. Sketch each transformed graph. The curves are schematic.

(b) $y=-f(2x)$

b: Graph



b: Coordinates

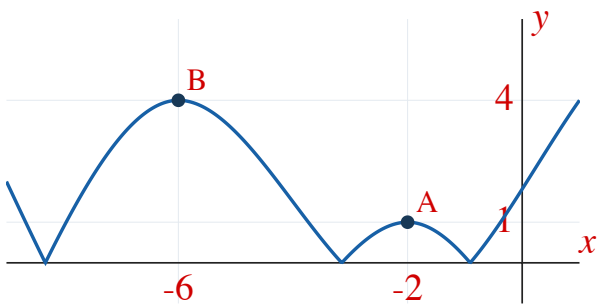
$(x,y)\mapsto(\frac{1}{2}x,-y)$.

Halve horizontal coordinates; negate vertical coordinates.

$A\to(1,1),\quad B\to(3,-4)$.

(c) $y=|f(-x)|$

c: Graph



c: Coordinates

$(x,y)\mapsto(-x,|y|)$.

Reflect in the vertical axis, then reflect negative outputs upwards.

$A\to(-2,1),\quad B\to(-6,4)$.

Test Your Understanding

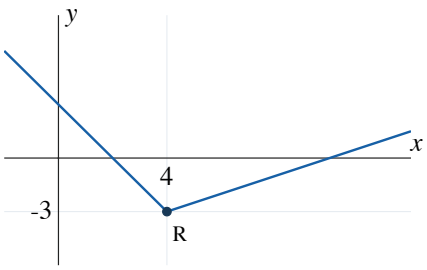
Solution 50

Edexcel C4 June 2011, Q3

The graph of $y=f(x)$ consists of two straight branches meeting at $R(4,-3)$. Sketch (a) $y=2f(x+4)$, (b) $y=|f(-x)|$. Show the point corresponding to R. (3 marks each)

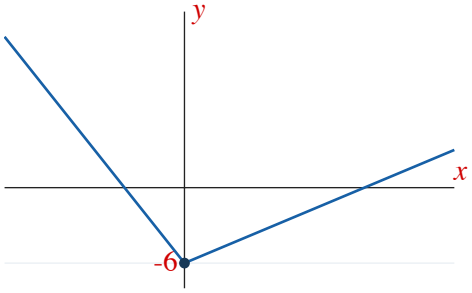
Diagrams are schematic.

Given graph



(a) $y=2f(x+4)$

a: Graph



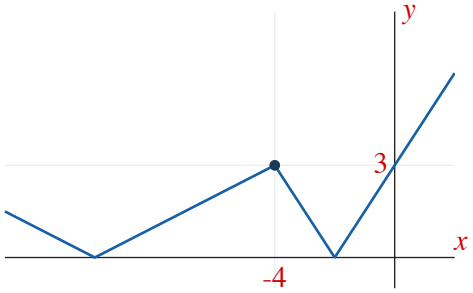
a: Point

$(x,y)\mapsto(x-4,2y)$, so

$R\to(0,-6)$.

(b) $y=|f(-x)|$

b: Graph



b: Point

$(x,y)\mapsto(-x,|y|)$, so

$R\to(-4,3)$.

What if two x changes or two y changes?

Solution 51

Two changes to one coordinate

For $y = 2f(x) + 1$, vertical coordinates become $2y + 1$: multiply by 2, then add 1.

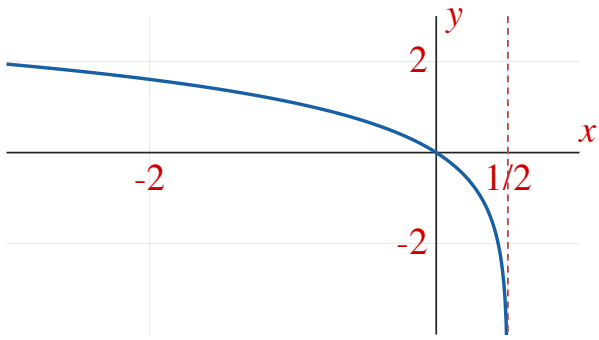
For $y = f(2x + 1)$, solve $2x_{\text{new}} + 1 = x_{\text{old}}$:

$x_{\text{new}} = \dfrac{x_{\text{old}} - 1}{2}$.

Subtract 1, then halve. The order matters!

Sketch $y = \ln(1 - 2x)$

Graph



Details

From $y = \ln x$, horizontal coordinates map as $x \mapsto \dfrac{1 - x}{2}$.

Reflect and halve horizontally, then translate $\frac{1}{2}$ right.

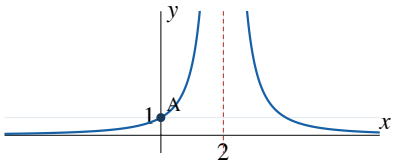
Domain: $x < \frac{1}{2}$. Vertical asymptote: $x = \frac{1}{2}$. Intercept: $(0, 0)$. The graph is decreasing.

Exercise 2F – Y2 Pages 47-48

Solution 52

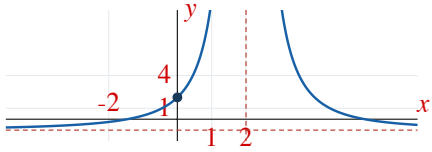
Exercise 2F, Q3

The schematic graph has asymptotes $x = 2$, $y = 0$, and passes through $A(0, 1)$. Sketch each transformation. State the asymptotes and the image of A.



(a) $y = 3f(x) - 1$

a: Graph



a: Details

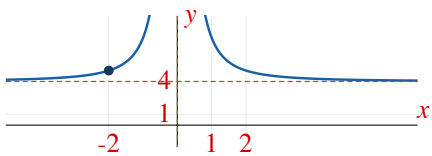
Stretch vertically by 3, then move down 1.

Asymptotes: $x = 2,\ y = -1$.

$A \to (0, 2)$.

(b) $y = f(x + 2) + 4$

b: Graph



b: Details

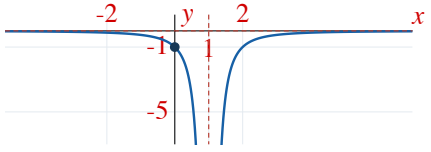
Translate 2 left and 4 up.

Asymptotes: $x = 0,\ y = 4$.

$A \to (-2, 5)$.

(c) $y = -f(2x)$

c: Graph



c: Details

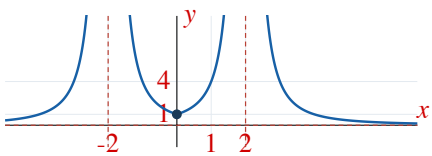
Halve horizontal coordinates; reflect in the horizontal axis.

Asymptotes: $x = 1,\ y = 0$.

$A \to (0, -1)$.

(d) $y = f(|x|)$

d: Graph



d: Details

Mirror the right half in the vertical axis.

Asymptotes: $x = -2,\ x = 2,\ y = 0$.

A stays at $(0, 1)$.

Exercise 2F – Y2 Pages 47-48 Solution 53

Exercise 2F, Q4

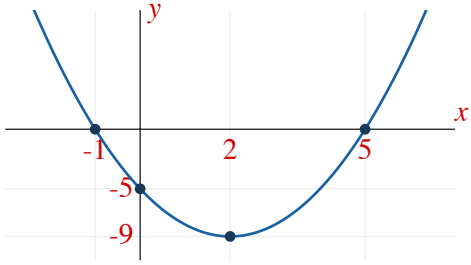
$g(x) = (x-2)^2 - 9,\ x \in \mathbb{R}$.

(a) Sketch g, labelling the turning point and intercepts.

(b) Find the image of its turning point under (i) $2g(x-4)$, (ii) $g(2x)$, (iii) $|g(x)|$.

(c) Sketch $g(|x|)$, labelling all turning points and intercepts.

a: Graph



b

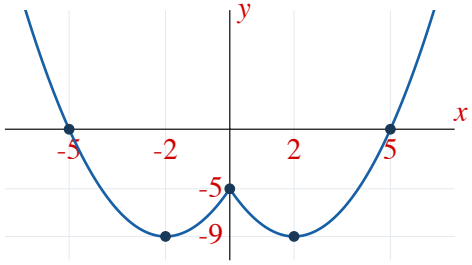
(i) $(2,-9) \mapsto (6,-18)$.

(ii) $(2,-9) \mapsto (1,-9)$.

(iii) $(2,-9) \mapsto (2,9)$.

In (iii), there are also sharp minima at $(-1,0)$ and $(5,0)$.

c: Graph



a: Points

Vertex: $(2,-9)$. Roots: $-1, 5$.

Vertical intercept: -5.

c: Points

Minima: $(-2,-9)$, $(2,-9)$.

Sharp local maximum: $(0,-5)$.

Roots: $-5, 5$.

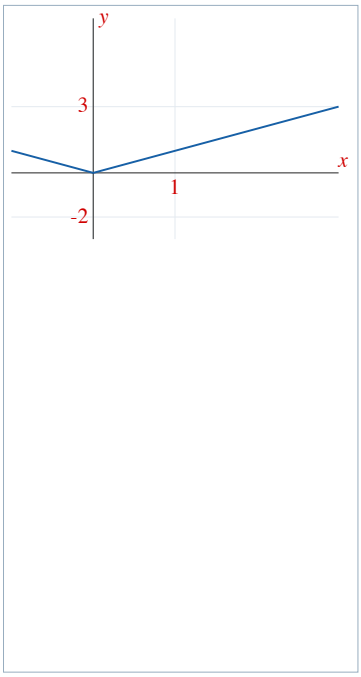
Solving Modulus Problems Solution 54

Given $f(x) = 3|x-1| - 2,\ x \in \mathbb{R}$:

(a) Sketch the graph. (b) State its range. (c) Solve $f(x) = \frac{1}{2}x + 3$.

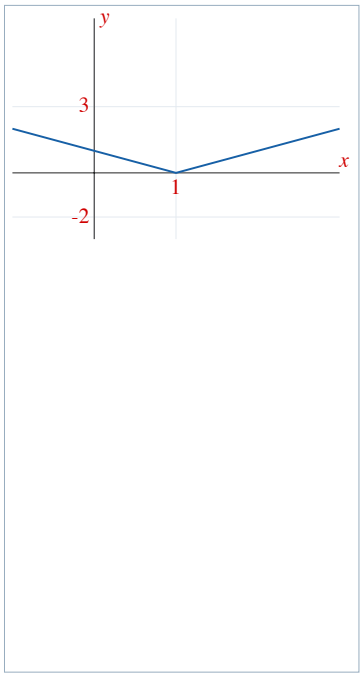
Part (a): build the graph in four stages.

$y = |x|$



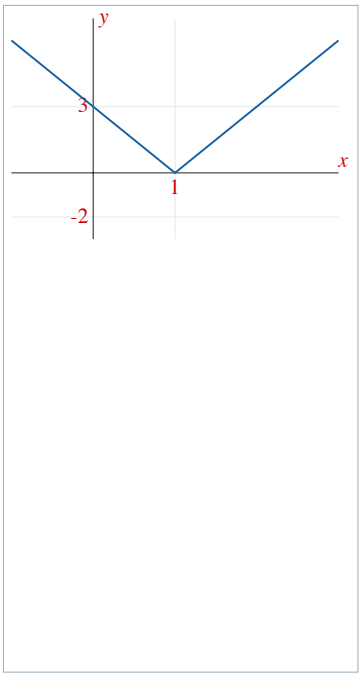
Start with the basic modulus graph.

$y = |x-1|$



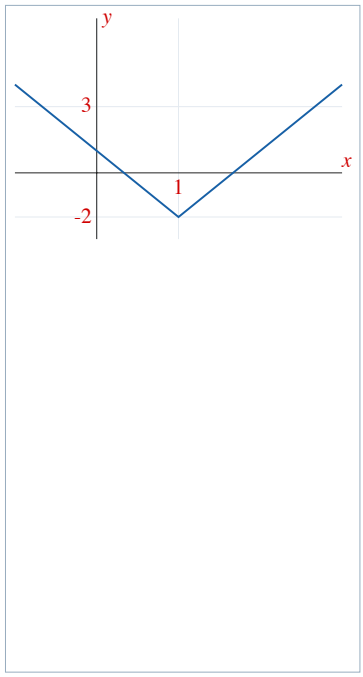
Translate 1 unit right.

$y = 3|x-1|$



Stretch vertically by factor 3.

$y = 3|x-1| - 2$



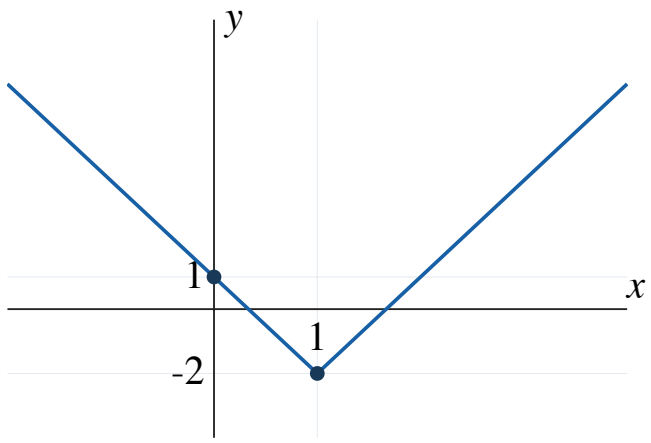
Translate 2 units down. Vertex: $(1,-2)$.

Solving Modulus Problems

Solution 55

$f(x) = 3|x-1| - 2,\ x \in \mathbb{R}.$

(b) State the range of f.



b

As $|x-1| \geq 0$,

$3|x-1| - 2 \geq -2.$

Equality holds at $x = 1$.

Therefore the range is

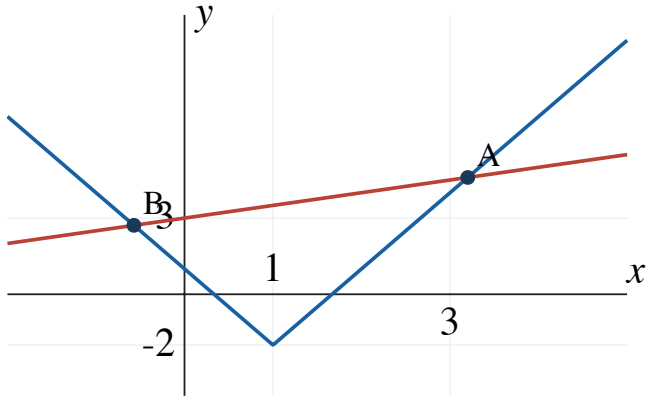
$[-2, \infty), \quad$ or $f(x) \geq -2.$

Solving Modulus Problems

Solution 56

$f(x) = 3|x-1| - 2.$

(c) Solve $f(x) = \frac{1}{2}x + 3$.



Blue: $f(x)$. Red: $\frac{1}{2}x + 3$. Split into branches at $x = 1$.

c: Right branch

For $x \geq 1$:

$$3(x-1) - 2 = \tfrac{1}{2}x + 3,$$
$$\tfrac{5}{2}x = 8,$$
$$x = \frac{16}{5}.$$

This satisfies $x \geq 1$.

c: Left branch

For $x < 1$:

$$3(1-x) - 2 = \tfrac{1}{2}x + 3,$$
$$-\tfrac{7}{2}x = 2,$$
$$x = -\frac{4}{7}.$$

This satisfies $x < 1$. Both are valid.

Test Your Understanding

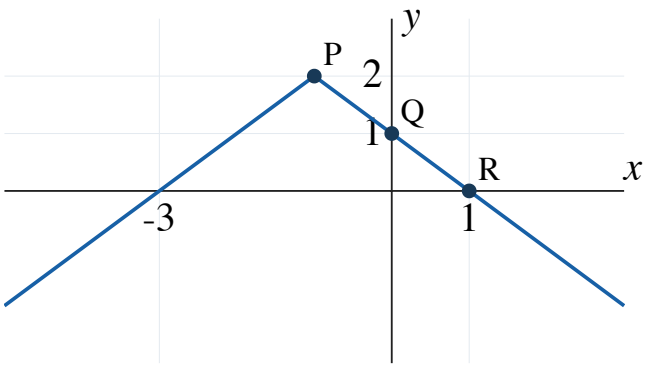
Solution 57

Edexcel C4 June 2008, Q3

Given $f(x) = 2 - |x + 1|$:

(c) Find P, Q and R. (3 marks)

(d) Solve $f(x) = \frac{1}{2}x$. (5 marks)



c

At the maximum, $|x + 1| = 0$, so $P = (-1, 2)$.

At $x = 0$, $f(0) = 1$, so $Q = (0, 1)$.

Solving $|x + 1| = 2$ gives roots $-3, 1$. The right intercept is $R = (1, 0)$.

d

For $x \geq -1$:

$2 - (x + 1) = \frac{1}{2}x \Rightarrow x = \frac{2}{3}$.

For $x < -1$:

$2 + (x + 1) = \frac{1}{2}x \Rightarrow x = -6$.

Both satisfy their branch conditions and the original equation.

Exercise 2G – Y2 Page 52

Solution 58

Exercise 2G, Q6

$m(x) = -2x + k, \qquad n(x) = 3|x - 4| + 6.$

Both have domain $\mathbb{R}$. Find the values of k for which $m(x) = n(x)$ has no real roots.

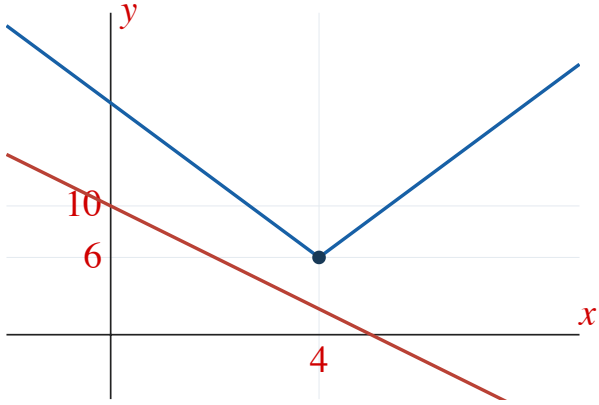
Answer

The line has slope -2, between the V-shaped graph's slopes -3 and 3. There are no intersections when the line is below the vertex at $x = 4$:

$-8 + k < 6 \Rightarrow \boxed{k < 14}$.

At $k = 14$ there is one root; for $k > 14$ there are two.

Example graph



Example shown: $k = 10$. The red line stays below the blue graph.

Exercise 2G, Q7

$s(x) = -10 - x, \qquad t(x) = 2|x + b| - 8.$

Both have domain $\mathbb{R}$. The equation $s(x) = t(x)$ has exactly one real root. Find b.

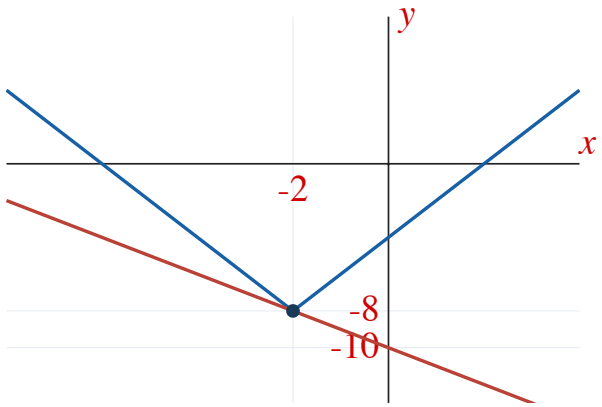
Answer

The line's slope -1 lies between the two branch slopes -2 and 2. Exactly one intersection occurs at the vertex $(-b, -8)$.

$-10 - (-b) = -8 \Rightarrow \boxed{b = 2}$.

The intersection is $(-2, -8)$.

Graph



Exercise 2G, Q8

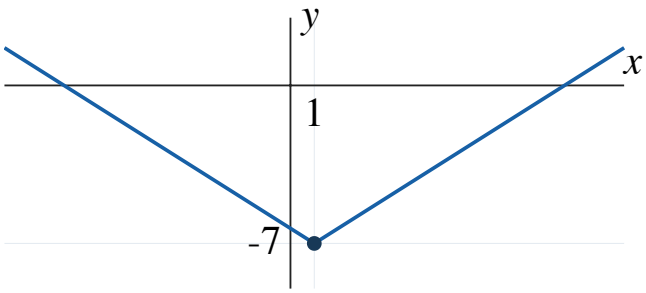
$h(x) = \frac{2}{3}|x - 1| - 7, \quad x \in \mathbb{R}.$

(a) State the range. (1 mark)

(b) Explain why h^{-1} does not exist. (1 mark)

(c) Solve $h(x) < -6$. (4 marks)

(d) Find k for which $h(x) = \frac{2}{3}x + k$ has no solutions. (4 marks)



a $h(x) \geq -7$. The minimum is at $x = 1$.

b h is many-to-one on $\mathbb{R}$, so it has no inverse function on this domain.

c

$\frac{2}{3}|x - 1| - 7 < -6 \Rightarrow |x - 1| < \frac{3}{2}$.

$-\frac{1}{2} < x < \frac{5}{2}$.

d

The right branch is $h(x) = \frac{2}{3}x - \frac{23}{3}$, parallel to the given line. No intersection occurs when

$$k < -\frac{23}{3}.$$

At equality the entire right branch coincides; above it there is one intersection on the left.

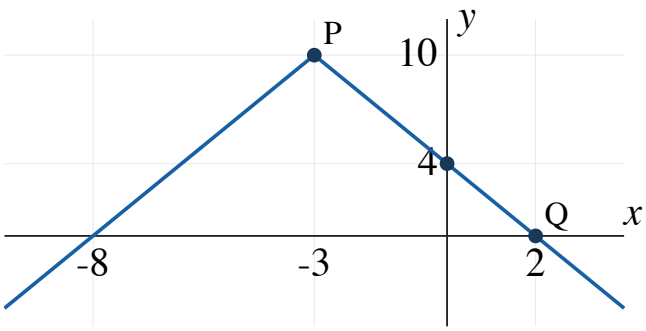
Exercise 2G, Q9

$h(x) = a - 2|x+3|$, $x \in \mathbb{R}$, crosses the vertical axis at $(0, 4)$.

(a) Find a. (2 marks)

(b) Find the maximum P and right-hand root Q. (3 marks)

(c) Solve $h(x) = \frac{1}{3}x + 6$. (5 marks)



a $4 = a - 2|3| = a - 6 \Rightarrow a = 10$.

b

The maximum occurs at $x = -3$, giving $P = (-3, 10)$.

$10 - 2|x+3| = 0$ gives roots $-8, 2$. Hence $Q = (2, 0)$.

c

For $x \geq -3$:

$4 - 2x = \frac{1}{3}x + 6 \Rightarrow x = -\frac{6}{7}$.

For $x < -3$:

$16 + 2x = \frac{1}{3}x + 6 \Rightarrow x = -6$.

Both satisfy their branch conditions.

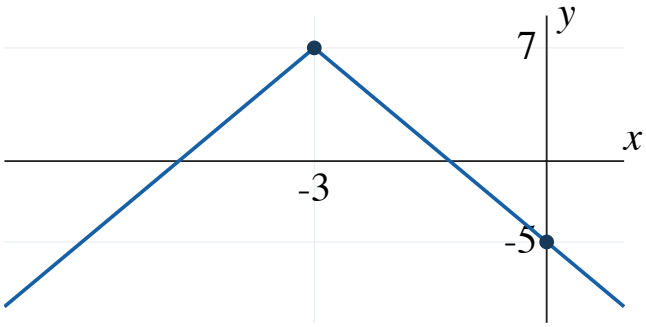
Exercise 2G, Q10

$m(x) = -4|x+3| + 7, \quad x \in \mathbb{R}$.

(a) State the range. (1 mark)

(b) Solve $m(x) = \frac{3}{5}x + 2$. (4 marks)

(c) Find the values of k for which $m(x) = k$ has two distinct roots. (4 marks)



a $m(x) \leq 7$, with maximum at $x = -3$.

b

For $x \geq -3$:

$-4x - 5 = \frac{3}{5}x + 2 \Rightarrow x = -\frac{35}{23}$.

For $x < -3$:

$4x + 19 = \frac{3}{5}x + 2 \Rightarrow x = -5$.

Both satisfy their branch conditions.

c

A horizontal line meets both branches exactly when it lies below the maximum:

$k < 7$.

MAT 2006, Q1I

How many solutions does the equation

$|x| + |x - 1| = 0$

have?

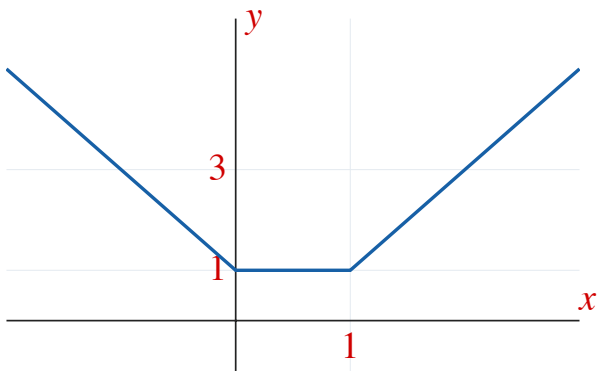
Answer

Both terms are non-negative. Their sum could be zero only if both were zero.

But $|x| = 0$ requires $x = 0$, whereas $|x - 1| = 0$ requires $x = 1$. They cannot both hold.

There are no solutions.

Graph



Alternatively, $|x| + |x - 1| \geq |x - (x - 1)| = 1$. The graph never reaches the horizontal axis.